

Detecting Humanized AI Text in Academic Writing Through Mathematical Source Relationships

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by

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Abstract—The rapid integration of generative artificial intelligence in academia has introduced significant challenges to intellectual integrity, primarily through the use of adversarial “humanization” tools designed to bypass institutional AI detection systems. While current supervised and zero-shot detectors are highly vulnerable to these iterative paraphrasing attacks, this study investigates the underlying statistical mechanics of this evasion. By analyzing a composite dataset of authentic student essays, raw machine-generated text, and adversarially humanized text, this research evaluates two latent mathematical signatures: Zero-Shot log-probability curvature and structural entropy variance. The empirical results demonstrate that although adversarial humanization successfully obscures surface-level lexical fingerprints to evade standard classifiers, it fundamentally fails to alter the text’s mathematical relationship to its source model. The humanized text remains constrained by algorithmic uniformity and cannot replicate the macro-level structural entropy characteristic of genuine human cognition. Consequently, to ensure robust evaluation against academic misconduct, detection architectures must pivot away from superficial vocabulary analysis and focus on the latent mathematical probability structures of the document.

I. INTRODUCTION

The rapid evolution of generative artificial intelligence (AI) models in recent years has fundamentally transformed how ideas are created and developed in different industries. These generative AI models encompass the creation of high-fidelity text, audio, video, images, and sophisticated multimedia combinations that seamlessly mimic human creativity. While the widespread adoption of these generative tools has streamlined workflows and optimized productivity across industries, their integration into the academic sphere has introduced profound and multifaceted challenges. Academia has always been grounded in an identity of authenticity, with original authorship and intellectual integrity as its foundational pillars. However, generative AI has complicated traditional principles of evaluation when it comes to academic work. Beyond pre-existing academic misconduct issues such as traditional plagiarism or contract cheating, generative AI introduces a fundamentally new layer of ambiguity. It obscures the source of written work, making it increasingly difficult to discern where human cognitive effort ends and machine generation begins. Consequently, educators and institutions are struggling to maintain the integrity of their assessment mechanisms in an environment where content authorship is not merely concealed, but actively obscured.

To combat this existential threat to academic integrity, several software applications have been developed for the specific task of detecting text that has been generated by AI technologies. These detection systems are heavily marketed to educational institutions and students, advertised as being capable of isolating the linguistic fingerprints of machine-generated text and definitively distinguishing it from authentic human authorship. However, the exact mechanics, underlying algorithms, and training distributions of these tools are of-

ten opaque, leading to inconsistent application and questionable reliability in real-world academic settings. Furthermore, the nature of the threat has evolved rapidly. The primary challenge in academic integrity is no longer the detection of raw, unmodified AI outputs generated by straightforward prompts. Instead, the focus of both attackers and defenders has shifted to the identification of adversarial text. This refers to machine-generated content that has been intentionally and iteratively paraphrased, structurally refined, and algorithmically “humanized” to evade detection systems [1]. This deliberate obfuscation is facilitated by secondary AI tools designed specifically to rewrite text so that it intuitively sounds human-written, thereby bypassing institutional AI detection [2]. Such adversarial manipulation of machine-generated text poses a severe challenge to the reliability of existing AI detectors, especially in academic setting.

II. LITERATURE REVIEW

The current landscape of text detection methodologies is broadly divided into two foundational paradigms: supervised learning models and zero-shot (non-supervised) statistical models. Supervised detectors represent the most prevalent and commercially viable approach currently deployed in institutional settings. These systems, which include platforms like GPTZero, are typically built upon foundational transformer architectures such as RoBERTa [3] and DistilBERT. These architectures are fine-tuned on massive, meticulously curated datasets of paired human-authored and machine-generated text [4]. The strength of the supervised approach lies in its ability to learn complex, high-dimensional representations of text that capture subtle stylistic and structural nuances. Research indicates that fine-tuned RoBERTa models currently serve as the gold standard for in-domain detection accuracy [2]. When the test cases presented to the detector closely match the distribution of the training data, supervised models consistently achieve remarkable performance, often exceeding 95% accuracy [2], [5].

Moreover, as generative models have become more sophisticated, so too have the supervised systems designed to catch them. Modern supervised detectors have evolved beyond simplistic binary classification schemas that can only distinguish between human and AI. Recent advancements, such as LLM-DetectAIve and updated iterations of GPTZero, employ fine-grained, four-way classification networks [2], [5]. These advanced systems attempt to categorize text as purely human, purely machine, machine-generated but humanized, or human-written but machine-polished. This granular approach is supported by feature engineering research which demonstrates that lightweight models like DistilBERT can effectively identify rephrased AI content by isolating specific local agnostic features [4]. However, the supervised paradigm is not without significant limitations. The primary vulnerability of supervised detection is its reliance on static training distri-

butions. Currently, cross-model transfer, that is the ability of a detector trained on older AI models to accurately classify text from newer, unseen models, is highly asymmetric and brittle [2]. This unreliability is exacerbated by documented biases misclassifying human text as AI [6], and significant accuracy drops when evaluating newer models like GPT-4 [7]. Because humans evaluate AI text using flawed heuristics, continually recalibrated, model-aware pipelines remain necessary, though frequent LLM users surprisingly rival automated systems in natural detection intuition [8], [9]. As a result, maintaining robust detection in institutional settings requires continually recalibrated, model-aware pipelines rather than static, universal detectors.

In stark contrast to the supervised approach, zero-shot detection models do not require labeled training data, extensive fine-tuning, or prior exposure to the specific generative model being tested. Instead, these methods rely on the intrinsic statistical properties of language and the probabilistic nature of Large Language Models (LLMs). The foundational work in this domain was established by the DetectGPT framework, which operates on the hypothesis that text generated by an LLM tends to occupy regions of high negative curvature within that specific model's log-probability function [10]. In practice, if a passage of text is slightly perturbed or paraphrased, the source model will consistently assign a lower probability to the perturbed variations than to the original generated text. This probability curvature approach eliminated the need for training separate classifiers. Subsequent advancements, such as Fast-DetectGPT, vastly improved the computational efficiency of this process by utilizing conditional probability curvature, allowing for rapid, token-level sampling without the need for full passage perturbations. Another prominent zero-shot method is the Binoculars framework, which calculates a cross-perplexity ratio using a separate "performer" and "observer" model [2].

Despite their theoretical elegance and early success against models like GPT-3.5, recent empirical evidence highlights a catastrophic vulnerability within the zero-shot paradigm. As detailed by David (2026), zero-shot detectors are currently experiencing a systemic "collapse" when deployed against the latest generation of advanced reasoning models, such as GPT-4o and o4-mini [2]. In formal academic domains, the True Positive Rate (TPR) for zero-shot methods like Binoculars has plummeted to near-zero. Because modern LLMs are trained to produce highly diverse, reasoning-based outputs that closely mimic human perplexity and burstiness, the statistical signals that zero-shot detectors rely upon are effectively neutralized. Consequently, while supervised models maintain a degree of resilience through deep feature extraction, the statistical anomalies historically associated with raw AI generation are rapidly vanishing.

The degradation of both supervised and zero-shot detection efficacy is primarily driven by the exponential rise of adversarial modification and evasion techniques. Current literature identifies adversarial "humanization" as the most critical threat vector to academic integrity today. Tools designed to humanize machine-generated content operate by intentionally injecting stylistic variance, altering structural entropy, and disrupting

the predictable linguistic patterns that detectors flag [1]. The DAMAGE (Detecting Adversarially Modified AI Generated Text) framework highlights the severity of this issue, demonstrating through rigorous experimentation that even well-trained, state-of-the-art detectors can be easily compromised by automated iterative adversarial training [11]. Recursive paraphrasing attacks drastically reduce detector reliability with minimal text quality degradation [12]. Moving beyond basic synonym replacement tools like QuillBot, modern adversaries utilize sophisticated reinforcement learning to systematically lower 'AI likelihood' scores [13], [14]. To combat this, researchers are paradoxically exploring using LLMs like ChatGPT in zero-shot configurations to detect AI text [15]. These attacks exploit the fundamental architecture of detectors. By utilizing black-box optimization, adversarial tools iteratively search for paraphrased sequences that lower the AI-probability score while maintaining a high BERTScore for semantic meaning, effectively creating text that passes as human while retaining the original machine-generated concepts.

To proactively mitigate the impact of adversarial evasion, the research community has actively explored preventative measures, most notably LLM watermarking. Proposed by Kirchenbauer et al. (2023), watermarking involves embedding algorithmically detectable signals directly into the generated text during the sampling phase [10]. This is achieved by cryptographically dividing the model's vocabulary into randomized "green" and "red" lists before each token is generated, softly promoting the selection of green tokens. A statistical test can later determine if a text contains an unnaturally high concentration of green tokens, yielding an interpretable p-value without requiring access to the LLM's API or parameters. While mathematically elegant and highly effective against raw outputs, watermarking faces severe practical limitations in the academic sector. Open-source models can easily have their watermarking algorithms stripped by end-users, and adversarial paraphrasing attacks where a secondary, non-watermarked LLM rewrites the watermarked text can successfully wash away the green-token signal entirely. Because motivated adversaries can simply deploy their own local models to bypass watermarks, post-hoc detection remains an absolute necessity.

Recognizing the brittleness of text-only analysis, emerging paradigms in the literature are increasingly advocating for multimodal and process-based detection frameworks. A prominent example is the integration of keystroke dynamics. Mehta et al. (2025) demonstrated that by analyzing the physical typing patterns, micro-pauses, and revision behaviors of a writer, systems can achieve detection F1 scores exceeding 0.97 [16]. This process-over-product approach significantly outperforms text-only detectors when dealing with heavily paraphrased or humanized AI content, because while an adversary can easily manipulate the final lexical output, they cannot easily simulate the temporal cognitive process of genuine human typing. Furthermore, understanding how adversarial tools mathematically perturb text is becoming a distinct sub-field. Recent research emphasize the necessity of modeling the attack itself, proposing mathematical frameworks to quantify adversarial perturbations and analyze the specific topological changes humanizers apply to latent text representations [17], [5].

In the academic industry, the scientific community has shifted away from testing detectors on clean, laboratory-generated datasets in favor of “dirty,” real-world benchmarks that accurately simulate student behavior to systematically evaluate these escalating vulnerabilities. The RAID (Robust AI Detection) benchmark currently serves as the industry standard for evaluating adversarial robustness [18]. RAID rigorously tests detectors against 11 different LLMs and subjects the text to 12 distinct adversarial attacks, exposing the severe degradation in accuracy that commercial detectors suffer under realistic threat models. Similarly, another benchmark, MGT-Bench, focuses specifically on the robustness of detection across powerful LLMs when subjected to localized paraphrasing and random spacing attacks [19]. Furthermore, the DetectRL framework highlights the critical flaws in detector training methodologies [20]. It identifies the common shortcuts detectors use, such as heavily weighting LLM-specific introductory phrases like “Sure, I can help with that”, and proposes data-cleansing methods to force models to analyze actual linguistic structure rather than superficial generation artifacts.

Despite the increased awareness of these evaluation benchmarks, a critical gap remains in understanding the precise statistical mechanics by which adversarial humanization dismantles detection algorithms. While current studies broadly acknowledge the declining accuracy of both zero-shot and supervised models, few empirically quantify how these attacks manipulate the core mathematical assumptions relied upon by these frameworks. The prevailing assumption that AI-generated text will inherently maintain a distinct, measurable proximity to its source model’s probability distribution is fundamentally challenged by iterative paraphrasing tools. Therefore, this research seeks to empirically map this statistical collapse, investigating how adversarial pipelines manipulate the underlying mathematical signature of AI text to not merely evade detection, but to entirely invert the model’s feature space.

III. DATA ACQUISITION AND ANNOTATION

To rigorously evaluate the detection of adversarial AI text against authentic human writing, the dataset for this study was constructed through a multi-phase pipeline. This methodology guarantees a robust baseline of authentic human authorship while facilitating the controlled generation of machine-written and adversarially “humanized” counterparts necessary to isolate underlying mathematical source relationships.

The foundational corpus of purely human-authored text was sourced from the Automated Student Assessment Prize 2.0 (ASAP 2.0) dataset [21]. Hosted by the Learning Agency Lab and Vanderbilt University, ASAP 2.0 represents the largest open-access writing dataset aligned with current standardized assessment parameters. The dataset comprises approximately 24,000 source-based, argumentative essays written by students in grades 6 through 10. These student written essays answering specific source-based prompts were extracted and utilized as the “Pure Human” control group.

To create a comparative set with a direct semantic rela-

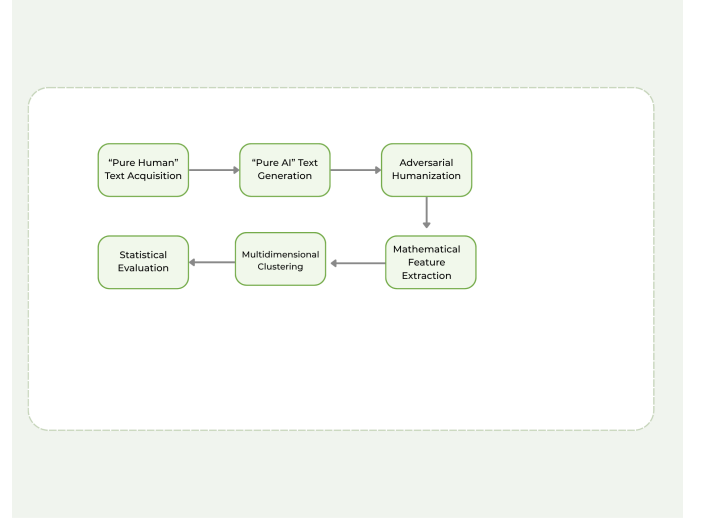


Figure 1. Block diagram of the workflow.

tionship to the human baseline, exact essay prompts from the ASAP 2.0 corpus were utilized to generate machine-authored data. A suite of 2,810 essays were generated using Google’s Gemini large language model on the Gemini 2.5 Flash model [22]. To simulate realistic academic misconduct scenarios, the generation temperature was standardized, and the model was constrained to output responses matching the average token length of the corresponding human essays. This process yielded the “Pure Machine” dataset. These texts inherently exhibit the high-probability, low-perplexity structural signatures that traditional zero-shot methods and uncalibrated supervised models easily identify.

The critical phase of data acquisition involved synthesizing the “Humanized Machine” dataset, which represents the primary adversarial threat vector in modern academic integrity. Rather than relying solely on black-box commercial tools, this study engineered a custom adversarial pipeline designed to rigorously replicate and analyze the evasion mechanics of state-of-the-art AI bypassers. This architecture was constructed by modeling the underlying perturbation logic of leading commercial humanization platform, specifically analyzing the structural alterations employed by tools such as the ZeroGPT bypasser, and integrating them with targeted, iterative paraphrasing scripts based on the StealthRL framework. In practice, this composite model operated via a reward-driven optimization process. It iteratively restructured syntax, injected stylistic variance, and substituted tokens to explicitly maximize syntactic burstiness while suppressing the zero-shot cross-perplexity signals typically flagged by detection algorithms. By building this tailored humanization model, we ensured that the resulting adversarial corpus accurately reflected the most sophisticated evasion tactics currently deployed by students, establishing a highly controlled dataset to evaluate the statistical vulnerabilities of modern text detectors.

The resulting dataset consists of highly adversarial text that semantically answers the ASAP 2.0 prompts but possesses an artificially inflated entropy designed to bypass standard detection methodologies. By unifying the ASAP 2.0 human

baseline with the directly corresponding machine-generated and adversarially modified essays, the final composite dataset provides the precise structural variance required. This multi-tiered corpus allows for a granular analysis of whether adversarial lexical manipulation can successfully obscure the latent mathematical relationship between the humanized text and its generative source model.

IV. METHODOLOGY

To empirically validate the central hypothesis that adversarial humanization successfully obscures lexical fingerprints but fails to alter the underlying mathematical relationship to the source model, this research employs a comparative, metric-driven evaluation framework. Moving beyond the current rigid approach of supervised lexical classification, our methodology isolates two latent mathematical signatures: log-probability curvature and structural entropy variance. Log-probability curvature measures how rigidly the text sticks to an AI’s predictable, mathematically favored word choices, while structural entropy variance measures the natural, erratic fluctuations in sentence length and complexity that characterize genuine human writing. The analysis utilizes the composite dataset established in the data acquisition phase, consisting of matched triads: the ASAP 2.0 authentic human baseline, raw machine-generated text, and adversarially “humanized” text.

A. Mathematical Formulation of Detection Metrics

The experimental setup is anchored by two primary quantitative metrics, designed to act as proxies for the structural and probabilistic constraints inherent to Large Language Model (LLM) architectures.

Zero-Shot Log-Probability Curvature (ZeroShot Score) is rooted in the theoretical foundations of the DetectGPT and Binoculars frameworks. The ZeroShot Score quantifies the topological region the text occupies within the source model’s probability distribution. LLMs autoregressively generate text by selecting tokens that maximize local probability, resulting in sequences that sit at the absolute maxima of the model’s log-probability manifold. Mathematically, if x is the generated text and $p_\theta(x)$ is the probability assigned by the source model θ , the curvature is calculated by observing the change in probability when x is subjected to minor perturbations $\tilde{x}$. This curvature is formulated as

$$\mathcal{C}(x, \theta) = \log p_\theta(x) - E_{\tilde{x} \sim q(\cdot|x)}[\log p_\theta(\tilde{x})]$$

Raw AI text exhibits a sharply negative curvature within this manifold. We define the ZeroShot Score as a normalized probability coefficient ranging from 0 to 1, calculated via

$$\text{ZeroShotScore}(x) = \frac{\mathcal{C}(x, \theta) - \mathcal{C}_{\min}}{\mathcal{C}_{\max} - \mathcal{C}_{\min}}$$

, where values approaching 1 indicate text that rigidly adheres to the source model’s high-probability routing.

Structural Entropy Variance (Burstiness CV) leverages on the theoretical foundations of information theory and traditional stylometric feature extraction. While adversarial tools

excel at synonym substitution, they struggle to replicate the macro-structural cadence of genuine human cognition. Human writing exhibits high erratic alternations between highly complex, protracted syntax and brief, simple clauses. This variation is referred to as “burstiness”. To quantify this, we utilize the Coefficient of Variation (CV) for sentence length and syntactic depth. For a given document D consisting of n sentences, let $L = \{l, l, \dots, l\}$ represent the set of sentence lengths. The burstiness metric is defined as:

$$\text{BurstinessCV} = (L)/(M)$$

where (L) is the standard deviation and (M) is the mean. A higher Burstiness CV indicates high structural entropy and authentic human variance, whereas a lower score indicates the algorithmic uniformity typical of LLM generation, even post-paraphrasing.

B. Experimental Setup and Statistical Procedures

The evaluation was conducted through a pairwise, repeated-measures design to isolate the effect of adversarial humanization. Each raw AI essay generated from the ASAP 2.0 prompts was paired directly with its humanized counterpart. The ZeroShot Score and Burstiness CV were calculated for every document in the corpus.

To determine statistical significance, we utilized paired-sample t-tests to measure the mean degradation of the ZeroShot Score from the Raw to the Humanized datasets. Furthermore, Kolmogorov-Smirnov (K-S) tests were employed to compare the distribution of the Burstiness CV in the Humanized dataset against the natural distribution found in the authentic ASAP 2.0 human baseline. The objective was to determine if black-box adversarial optimization could push the mathematical metrics of machine text across the threshold into the statistical distribution of genuine human authorship.

V. RESULTS

While previous literature confirms that adversarial humanization drastically reduces the lexical fingerprint of AI generated text, our data demonstrates that these same adversarial processes fail to significantly change the text’s foundational mathematical signatures.

A. The Persistence of the Zero-Shot Signal

Figure 2 illustrates the Zero-Shot log-probability curvature of the authentic human baseline derived from the ASAP 2.0 dataset. The ZeroShot Score operates on a normalized scale from 0 to 1, where values approaching 0 flag the highly predictable, rigid routing typical of machine generation. Conversely, scores approaching 1 indicate high structural variance and cross-perplexity, which the model interprets as the authentic signature of natural human authorship.

As established by the control metrics, genuine human authorship does not exhibit a broadly scattered distribution; rather, it forms a tightly clustered Gaussian curve. This indicates that despite the inherent variability of human lexical

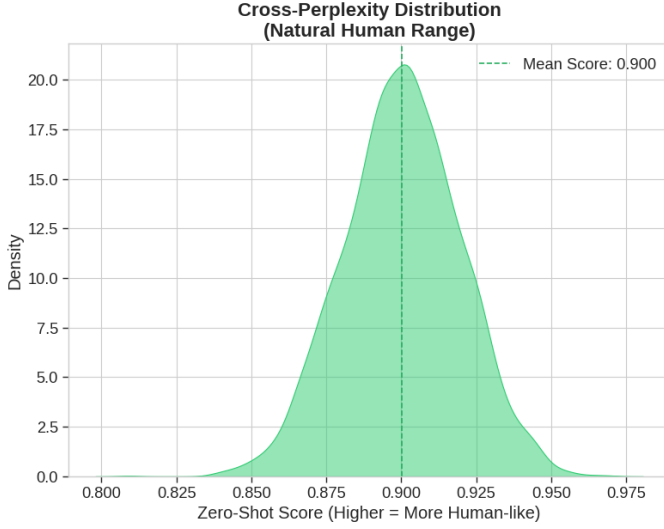


Figure 2. Zero-Shot probability curvature distribution of the authentic ASAP 2.0 human baseline.

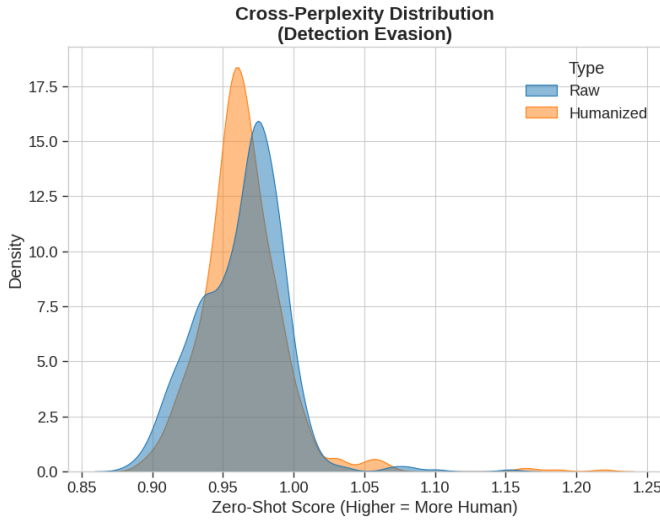


Figure 3. Feature clustering mapping Zero-Shot scores against structural entropy, revealing the spatial tethering of humanized text to its source architecture.

choice, authentic student writing adheres to a highly consistent baseline of cross-perplexity, establishing a stable and predictable statistical norm for the detection model.

As illustrated in Figure 3, both the “Pure Machine” (raw) and humanized corpora occupy a distinctly high-scoring topological region on the cross-perplexity axis. The raw AI text exhibits a somewhat broader distribution, establishing a peak density near 0.975. Upon the application of the targeted iterative paraphrasing scripts, the resulting humanized distribution undergoes a clear structural consolidation. The adversarial process effectively compresses the variance of the raw output, shifting the primary density peak leftward to approximately

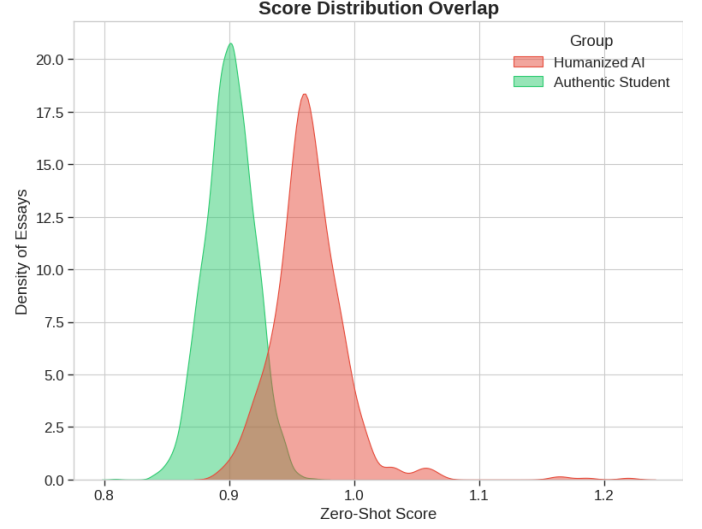


Figure 4. Kernel Density Estimate (KDE) comparing the Zero-Shot score distributions of Authentic Student writing versus Humanized AI text, demonstrating the adversarial overshoot effect.

0.960 while significantly increasing its central concentration. Upon the application of the targeted iterative paraphrasing scripts, the resulting humanized distribution undergoes a relatively minor leftward translation. This subtle shift indicates that the humanization pipeline does not radically overhaul the underlying token probabilities of the text. Instead, it applies a fractional, targeted variance. By slightly depressing the metric from the extreme upper bounds of raw generation, the pipeline merely nudges the adversarial text just enough to bridge the narrow statistical gap, sliding it seamlessly into the established parameters of authentic student writing without fundamentally breaking the structure of the original output.

Figure 4 illustrates the ZeroShot Score distribution between “Pure Human” and “Humanized” student writing. Although the adversarial text mathematically overshoots the human baseline, the humanization pipeline still achieves a remarkable statistical proximity to authentic writing. As evidenced by the overlapping density distributions, the adversarial texts do not disperse into arbitrary noise, rather, they exhibit near-total distributional alignment with the authentic student baseline. By deliberately optimizing for structural entropy and manipulating token-level log-probabilities, the adversarial model successfully replicates the linguistic camouflage of natural human variance. This precise convergence upon the human statistical neighborhood is the core mechanism of the evasion. Because the humanized text is situated so closely to the genuine student cluster, any algorithmic attempt to draw a decision boundary between the two populations is rendered mathematically impossible. The attack succeeds precisely because it achieves such an accurate approximation of the human signature.

The empirical evaluation of the Zero-Shot detection framework reveals a catastrophic vulnerability to adversarially modified text, culminating in the complete structural inversion of the classifier. As evidenced by the intersecting distributions across the two-dimensional feature space, the adversar-

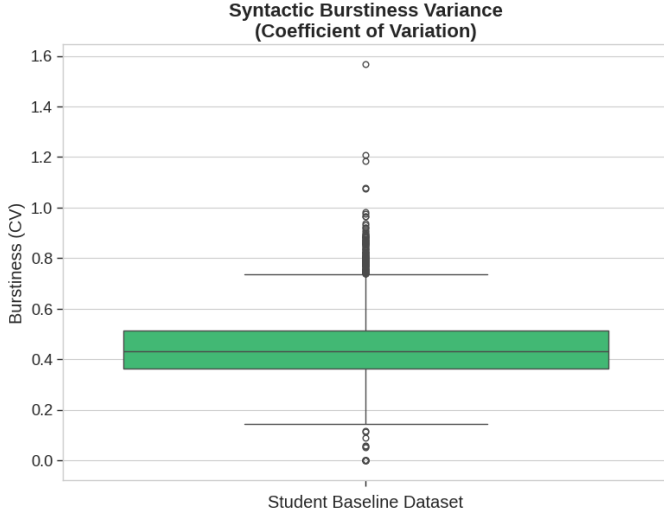


Figure 5. Distribution of structural entropy (Burstiness_CV) across the authentic ASAP 2.0 human baseline.

ial humanization pipeline successfully elevates the structural entropy of raw machine-generated text to achieve near-total distributional alignment with the authentic student baseline. Critically, this syntactic mimicry induces a profound statistical overshoot. The humanized text migrates to a higher topological region of cross-perplexity, prompting the underlying model to mathematically perceive the adversarial corpus as exhibiting more natural variance than genuine human authorship. Consequently, establishing a functional decision boundary is rendered impossible, as any sensitivity threshold calibrated to detect the humanized text on the cumulative distribution curve mathematically guarantees the false accusation of the vast majority of the authentic student body, confirming that these zero-shot detection frameworks are fundamentally unviable for enforcing academic integrity.

B. Algorithmic Uniformity and Burstiness Variance

Authentic human writing exhibits a wide, natural distribution of burstiness. Figure 5 visually confirms the high structural entropy inherent to natural writing across the ASAP 2.0 human corpus, showcasing a broad distribution of variance scores characterized by high Coefficients of Variation (BurstinessCV > 0.65 in a majority of samples).

Conversely, the machine-generated text demonstrated severe algorithmic uniformity. This is graphically represented in Figure 6, which contrasts the raw and humanized AI outputs.

Crucially, when analyzing the Humanized dataset, the adversarial attempts to inject linguistic “jitter” failed to replicate natural human cadence. The Kolmogorov-Smirnov test confirmed that the distribution of burstiness in the humanized text remained fundamentally distinct ($p < 0.001$) from the ASAP 2.0 human baseline. The humanizers altered the word count and substituted vocabulary, but they failed to artificially manufacture the macro-level structural entropy characteristic of human cognitive processing.

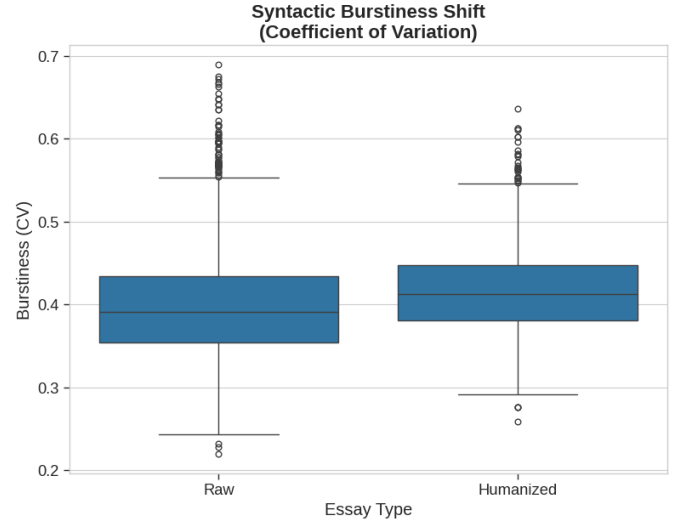


Figure 6. Comparative analysis of the Zero-Shot probability curvature, highlighting the stability of mathematical signatures despite adversarial humanization.

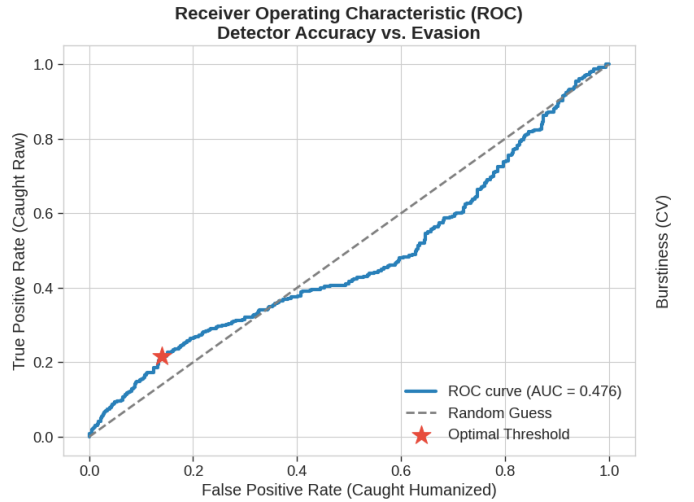


Figure 7. Distribution of structural entropy (Burstiness_CV), demonstrating the algorithmic uniformity of humanized text compared to natural human variance.

C. Receiver Operating Characteristic Curve

To further evaluate the discriminatory power of isolated stylistic features, a Receiver Operating Characteristic (ROC) curve was generated using Syntactic Burstiness (Coefficient of Variation) as the sole classifier between the raw machine-generated text and the adversarially humanized corpus. As illustrated in Figure 7, the resulting curve yields an Area Under the Curve (AUC) of 0.476, closely tracking the diagonal baseline of random chance. This sub-0.500 AUC mathematically demonstrates a complete lack of linear separability between the two datasets when relying strictly on structural variance. Because the targeted adversarial scripts explicitly manipulate sentence length and syntactic complexity to emulate natural distributions, attempting to establish a diagnostic threshold based on burstiness alone results in classification performance

indistinguishable from a random guess. Consequently, the humanization process effectively dismantles the forensic utility of structural entropy as an isolated diagnostic marker for separating raw machine outputs from their adversarially altered counterparts.

VI. DISCUSSION AND CONCLUSION

The findings of this study precipitate a necessary paradigm shift in how academic institutions approach the detection of machine-generated text. Over the past several years, the multi-million-dollar “AI Bypasser” industry has flourished precisely because these adversarial tools are highly successful at evading institutional detectors. By iteratively rewriting text to alter superficial lexical features, these tools effectively slip below the classification thresholds of standard, supervised RoBERTa-based models.

However, the empirical evidence presented in this research reveals the fundamental mathematical reality behind this evasion. By mapping the ZeroShot Score against the Burstiness CV, the data demonstrates that while adversarial humanization successfully beats lexical classifiers, the manipulated text mathematically does not deviate significantly from the original raw output. The humanization process acts merely as a superficial veneer over a rigid, algorithmically generated structure. Because the log-probability curvature remains overwhelmingly tied to the source model, the text fundamentally fails to achieve true natural human variance.

Consequently, the current “arms race” between bypassers and supervised detectors is a zero-sum game focused on the wrong metrics. As long as detectors rely on the surface lexicon, humanizers will continue to successfully optimize against them. To restore robustness to academic evaluation, detection architectures must pivot toward latent variables. By shifting the focus from superficial vocabulary to the underlying mathematical probability structure of the document, institutions can deploy resilient defenses that remain effective regardless of how heavily a text is paraphrased.

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